

Database Management System

CSIT — 4th Semester — 2081 — Answer Sheet
Subject Code: CSC265

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What do you mean by entity type and entity set? Construct an ER diagram for an airline ticket booking system where customers buy airline tickets online. The system should provide discounts on the basis of number of tickets bought. It should also keep records of buyers on the basis of number of times buyer visits the system. It should also filter the unwanted visitors to manage traffic congestion.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Consider the library database schema containing the tables given below where primary keys are underlined. USER (UserID, User_name, Address) USES (UserID, BookID, Issued_date, Returned_date) BOOK (BookID, Book_name, Author, PublisherID) PUBLISHER (PublisherID, Publisher_name, Address) Specify the following queries in both SQL and Relational Algebra on this database schema. a. Retrieve names of all books issued to the user "Hari". b. Retrieve the names of all books published by "ABC publication". c. Count the number of books published by the author "Ram".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are desirable properties of transactions? Explain conflict serializability with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. What do you mean by referential integrity? Why do you need this integrity?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain ANSI/SPARC three-schema architecture with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What are informal design guidelines for relational schemas.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Classify database management systems based on the data model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define the terms data, database, DBMS, database system, and database catalog.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain relational algebra natural join (*) operation with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain deferred update approach in database recovery.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain transaction deadlock with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Define specialization. Explain disjoint constraint with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.